
DESIGN

a design algorithm for any digital units (logic circuits), given truth table

Optimistic vs. Pessimistic minterms (SoP) vs. MAXTERMS (PoS)

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1. minterm

aka. Standard Product (ANDs)

Optimistic

X' vs. X

1 binary variable appear either:

- in its normal form X , or
- in its complement form X'

m_0

X'

m_1

X

YX vs. YX' vs. $Y'X$ vs. $Y'X'$

2 binary variables appear either in one of these forms:

m_0	$Y'X'$
m_1	$Y'X$
m_2	YX'
m_3	YX

ZYX vs. ZYX' vs. ...

3 binary variables appear either in one of these forms: how many?

ZYX vs. ZYX' vs. ...

3 binary variables appear either in one of these forms: how many?
Each variable can take 2 forms (normal and complement)
We have 3 variables, $2 \times 2 \times 2 = 2^3 = 8$

m_0	$Z'Y'X'$
m_1	$Z'Y'X$
m_2	$Z'YX'$
m_3	$Z'YX$
m_4	$ZY'X'$
m_5	$ZY'X$
m_6	ZYX'
m_7	ZYX

$$A_{n-1} \cdots A_2 A_1 A_0 \text{ vs. } A_{n-1} \cdots A_2 A_1 A'_0 \dots$$

n binary variables appear either in one of these forms: how many?

Each variable can take 2 forms (normal and complement)

We have n variables, $2 \times 2 \times 2 \times \cdots \times 2 = 2^n$

m_0	$A'_{n-1} \cdots A'_2 A'_1 A'_0$
m_1	$A'_{n-1} \cdots A'_2 A'_1 A_0$
m_2	$A'_{n-1} \cdots A'_2 A_1 A'_0$
$\vdots$	$\vdots$
$\vdots$	$\vdots$
m_{2^n-3}	$A_{n-1} \cdots A_2 A'_1 A_0$
m_{2^n-2}	$A_{n-1} \cdots A_2 A_1 A'_0$
m_{2^n-1}	$A_{n-1} \cdots A_2 A_1 A_0$

2. TRUTH TABLE

en.wikipedia.org/wiki/Truth_table

X	$F = F(X) = ?$
0	?
1	?

X	$F = F(X) = 0$
0	0
1	0

No Gate
No Circuit

X	$F = F(X) = X'$
0	1
1	0

m_0

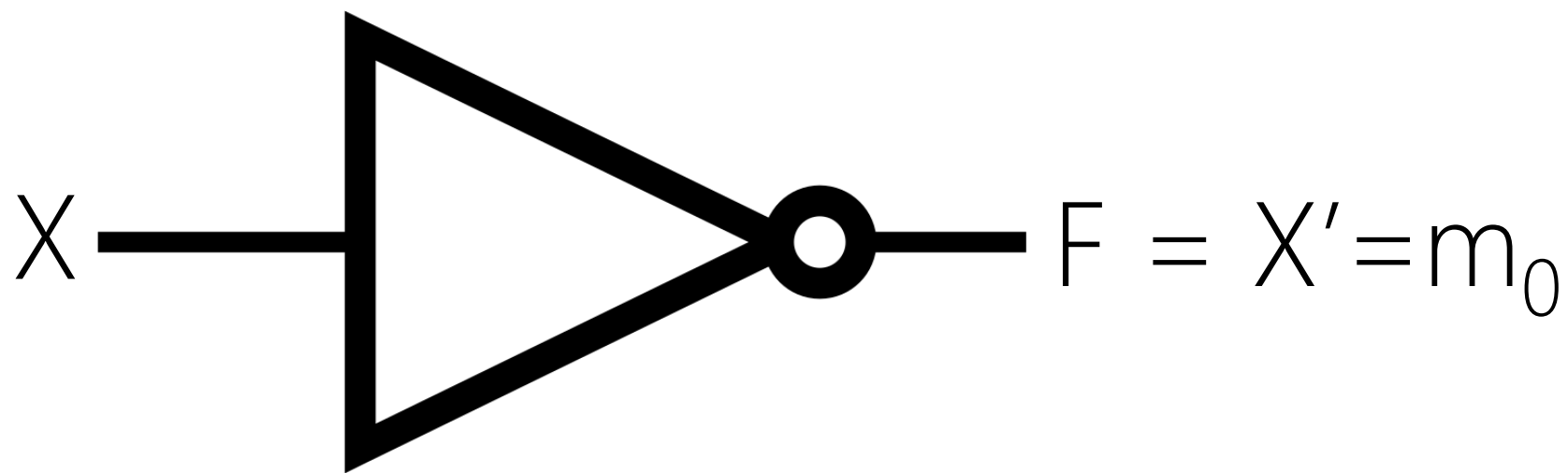
X'

m_1

X

X	$F = F(X) = X' = m_0$
0	1
1	0

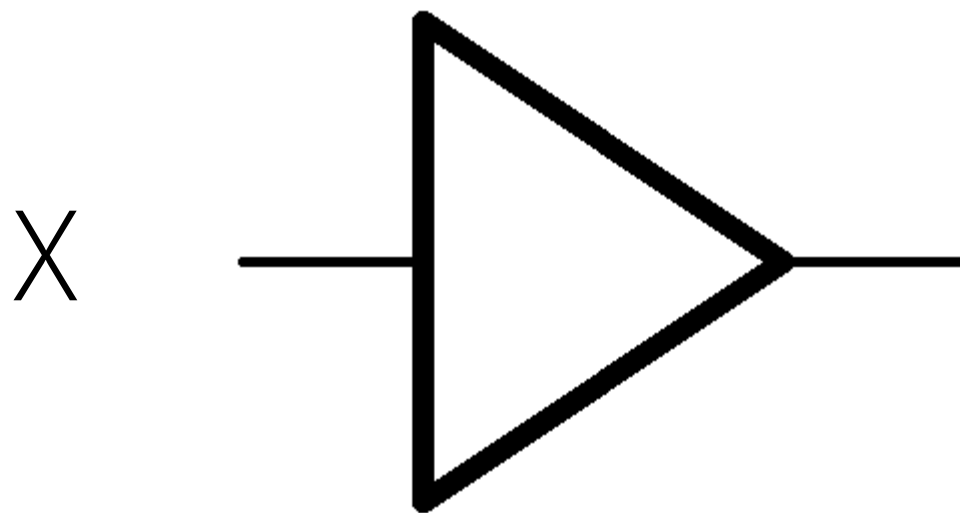
X	$F = F(X) = X' = m_0$
0	1
1	0



X	$F = F(X) = X$
0	0
1	1

X	$F = F(X) = X = m_1$
0	0
1	1

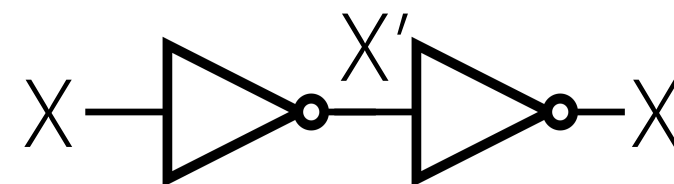
X	$F = F(X) = X = m_1$
0	0
1	1



X

$$F = X = m_1$$

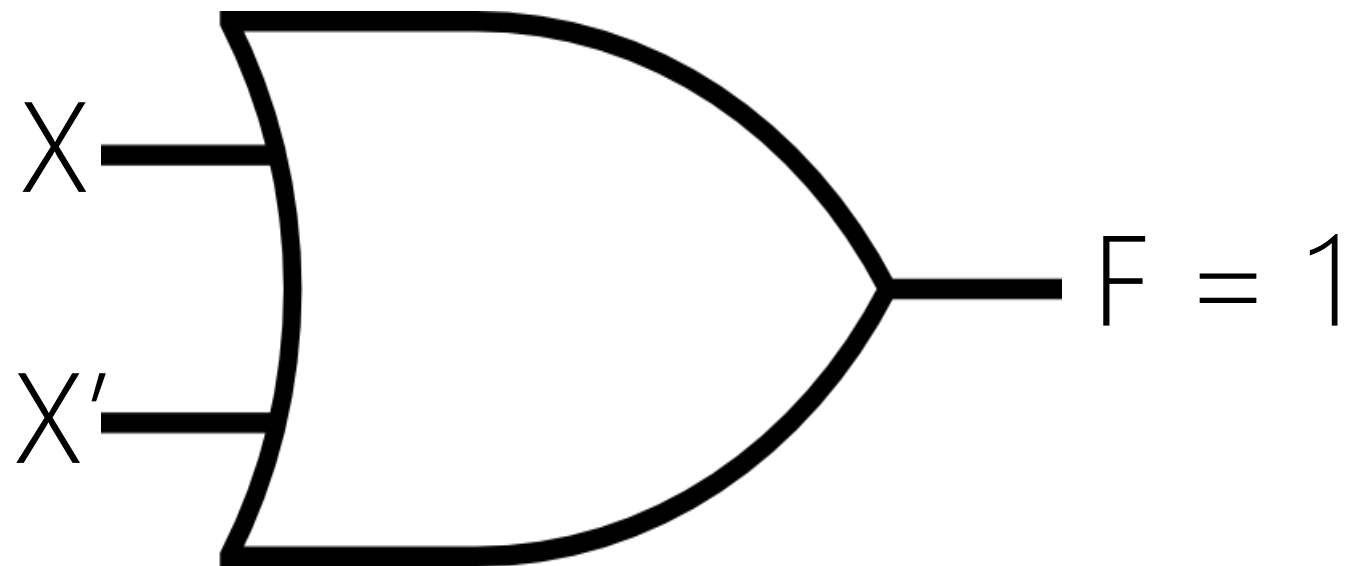
Digital Buffer



X	$F = F(X)$
0	1
1	1

X	$F = F(X) = X' = m_0$
0	1
1	1

X	$F = F(X) = X' + X = m_0 + m_1$
0	1
1	1



X'	X	X'+X
1	0	1
0	1	1

$$F = X + X' = 1$$

TRUTH TABLE $\leftrightarrow$ minterm

Y	X	$F = F(Y,X) = ?$
0	0	?
0	1	?
1	0	?
1	1	?

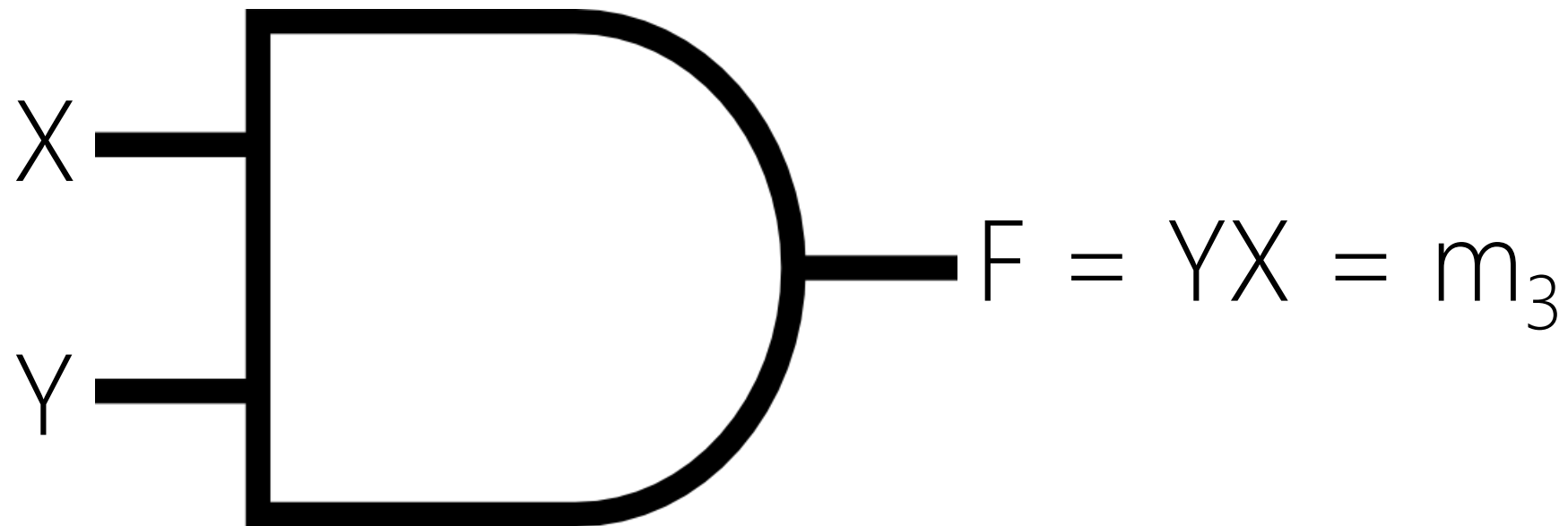
Y	X	$F = F(Y,X) = 0$
0	0	0
0	1	0
1	0	0
1	1	0

Y	X	$F = F(Y,X) = YX$
0	0	0
0	1	0
1	0	0
1	1	1

m_0	$Y'X'$
m_1	$Y'X$
m_2	YX'
m_3	YX

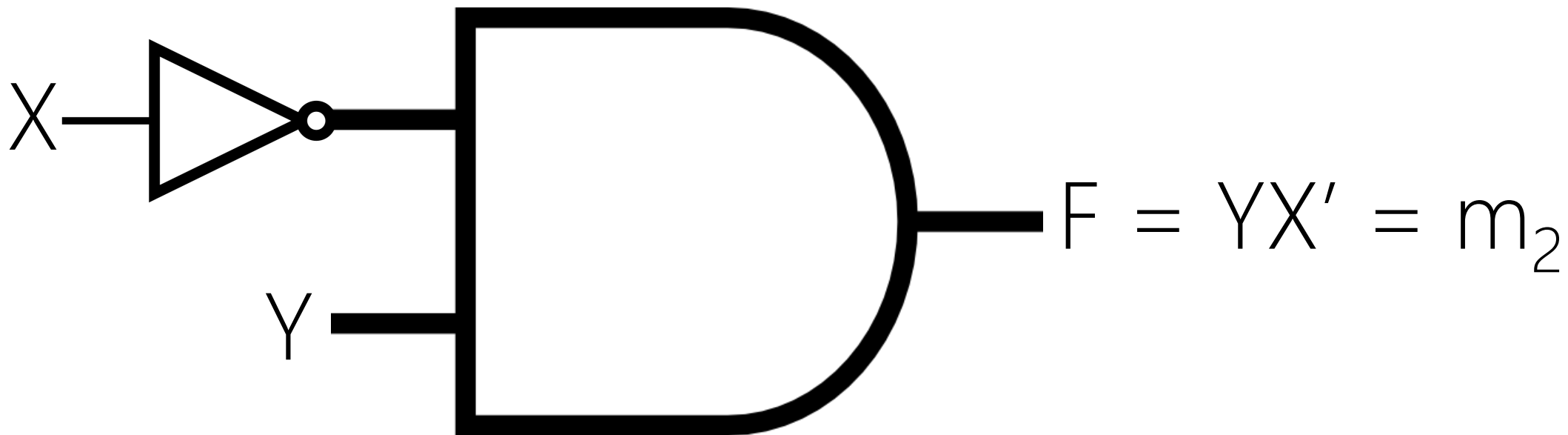
Y	X	$F = F(Y,X) = YX = m_3$
0	0	0
0	1	0
1	0	0
1	1	1

Y	X	$F = F(Y,X) = YX = m_3$
0	0	0
0	1	0
1	0	0
1	1	1



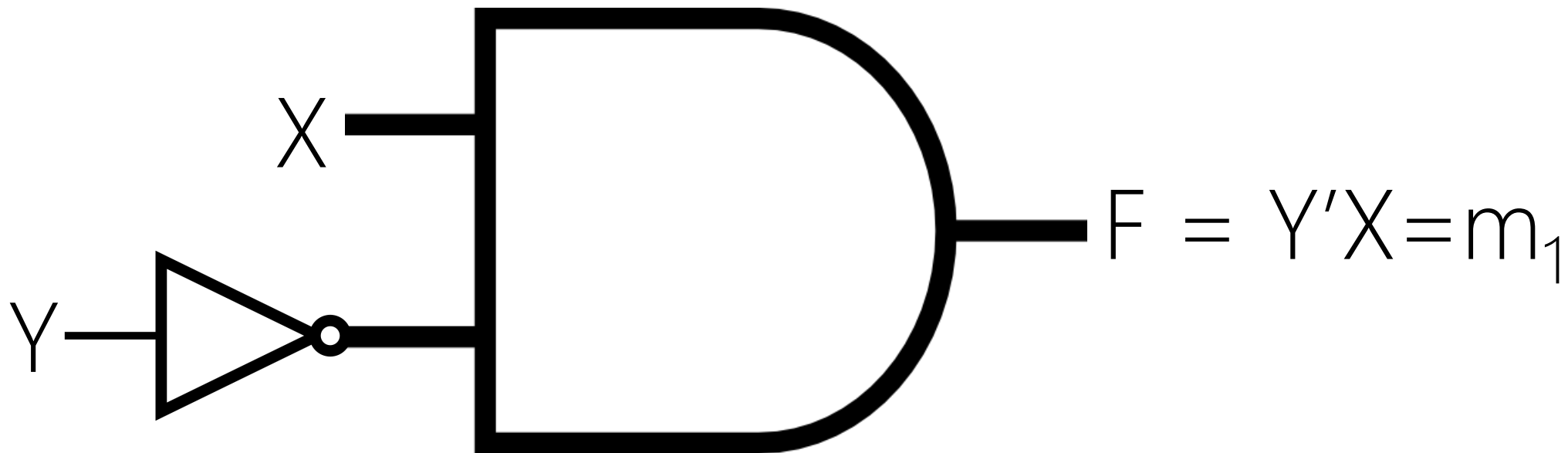
Y	X	$F = F(Y,X) = YX' = m_2$
0	0	0
0	1	0
1	0	1
1	1	0

Y	X	$F = F(Y,X) = YX' = m_2$
0	0	0
0	1	0
1	0	1
1	1	0



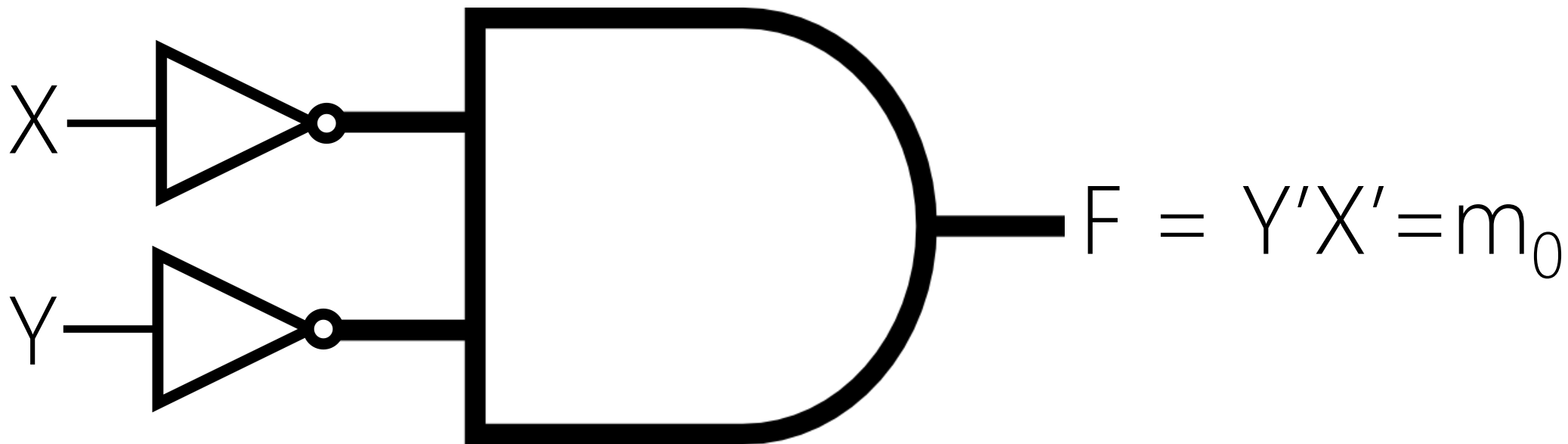
Y	X	$F = F(Y,X) = Y'X = m_1$
0	0	0
0	1	1
1	0	0
1	1	0

Y	X	$F = F(Y,X) = Y'X = m_1$
0	0	0
0	1	1
1	0	0
1	1	0



Y	X	$F = F(Y,X) = Y'X' = m_0$
0	0	1
0	1	0
1	0	0
1	1	0

Y	X	$F = F(Y,X) = Y'X' = m_0$
0	0	1
0	1	0
1	0	0
1	1	0



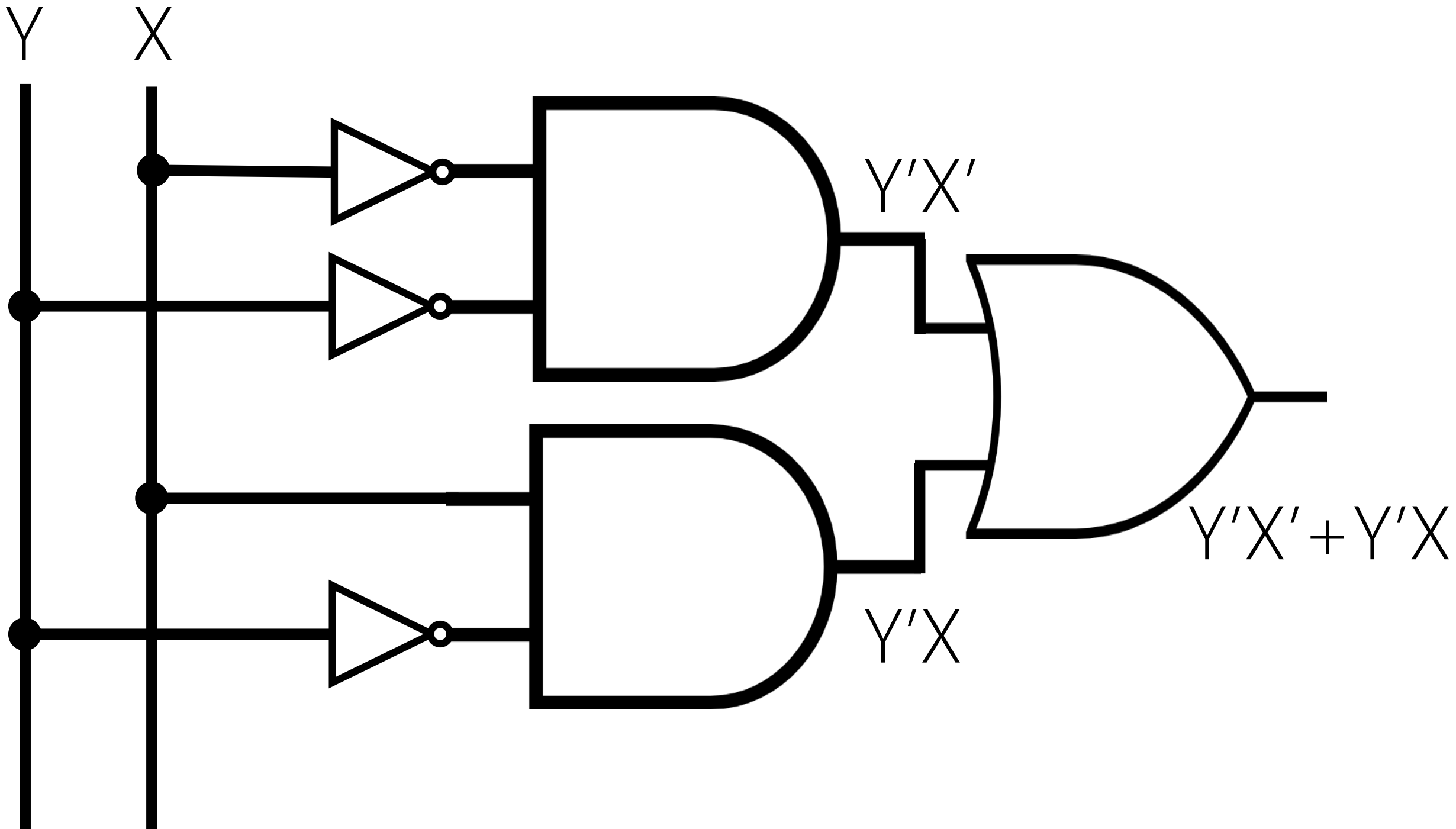
Y	X	$F = F(Y,X) = ?$
0	0	1
0	1	1
1	0	0
1	1	0

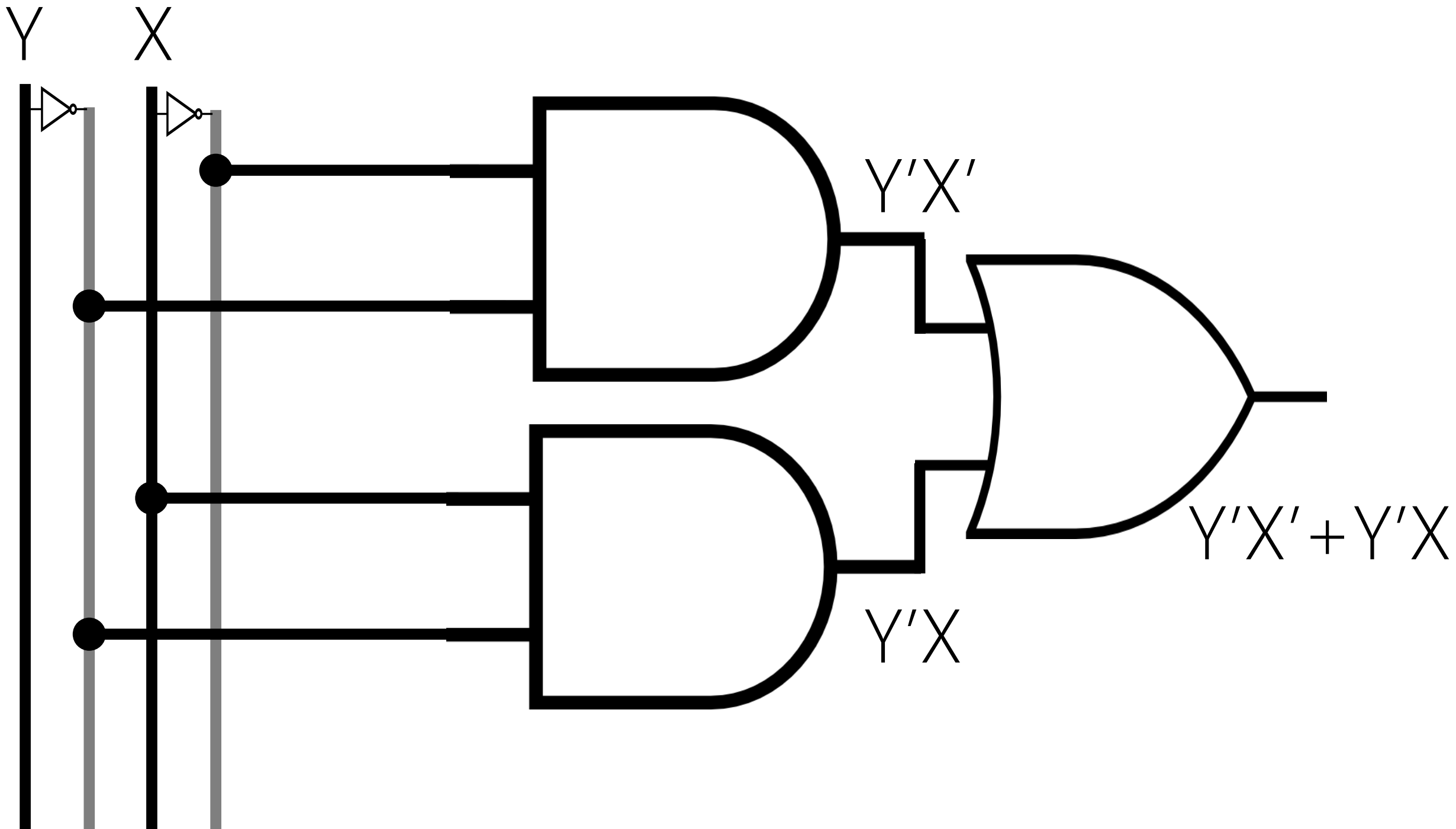
Y	X	$F = F(Y,X) = Y'X'$
0	0	1
0	1	1
1	0	0
1	1	0

Y	X	$F = F(Y,X) = Y'X' + Y'X$
0	0	1
0	1	1
1	0	0
1	1	0

Y	X	$F = F(Y,X) = m_0 + m_1$
0	0	1
0	1	1
1	0	0
1	1	0

Y	X	$F = F(Y,X) = \sum m(0,1)$
0	0	1
0	1	1
1	0	0
1	1	0





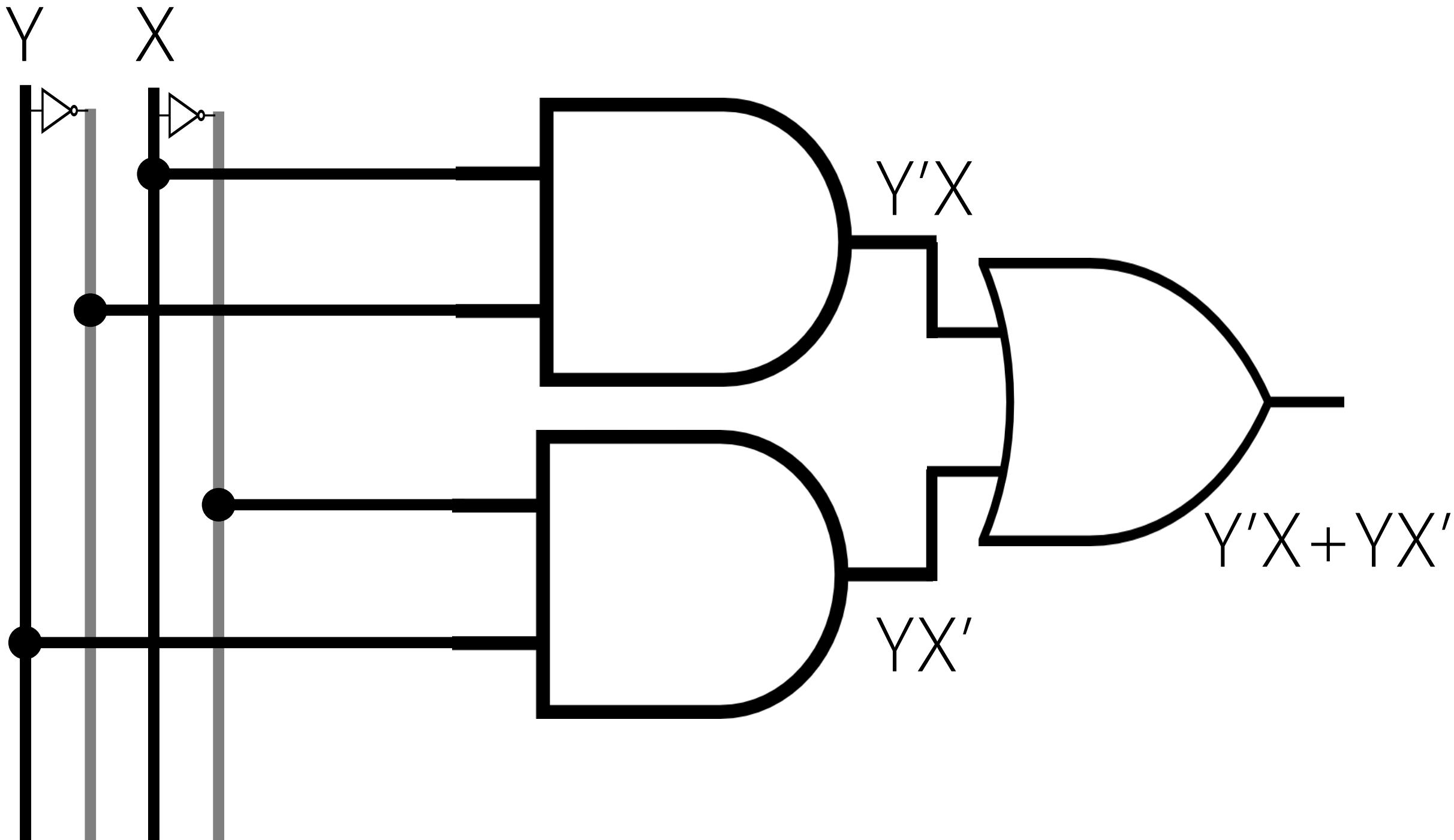
Y	X	$F = F(Y,X) = ?$
0	0	0
0	1	1
1	0	1
1	1	0

Y	X	$F = F(Y,X) = Y'X$
0	0	0
0	1	1
1	0	1
1	1	0

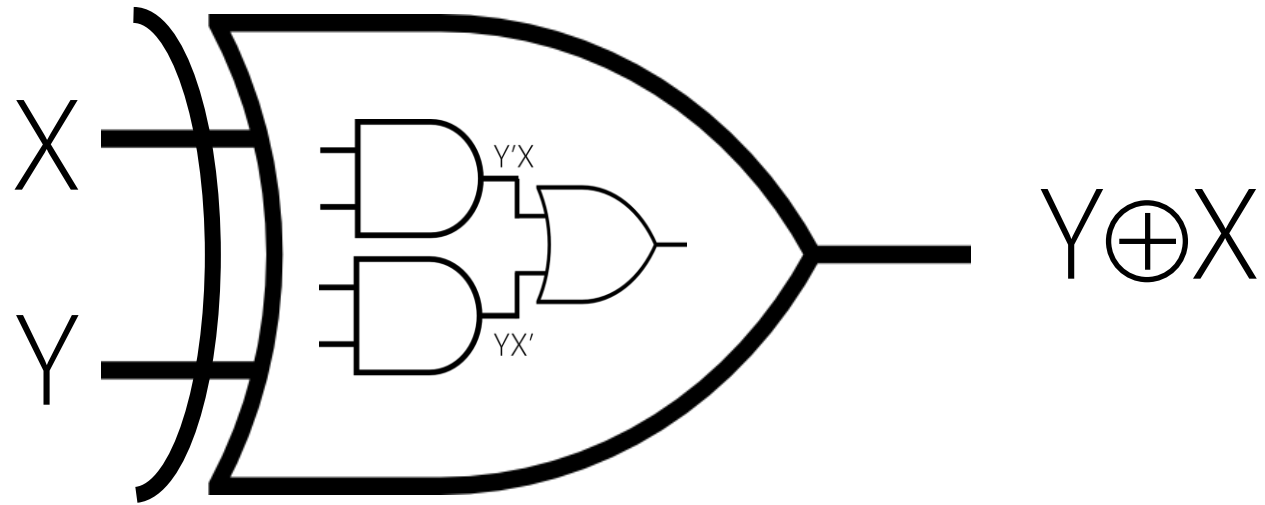
Y	X	$F = F(Y,X) = Y'X + YX'$
0	0	0
0	1	1
1	0	1
1	1	0

Y	X	$F = F(Y,X) = m_1 + m_2$
0	0	0
0	1	1
1	0	1
1	1	0

Y	X	$F = F(Y,X) = \sum m(1,2)$
0	0	0
0	1	1
1	0	1
1	1	0



Exclusive-OR (XOR)



Y	X	$F = F(Y,X) = Y'X + YX' = m_1 + m_2$
0	0	0
0	1	1
1	0	1
1	1	0

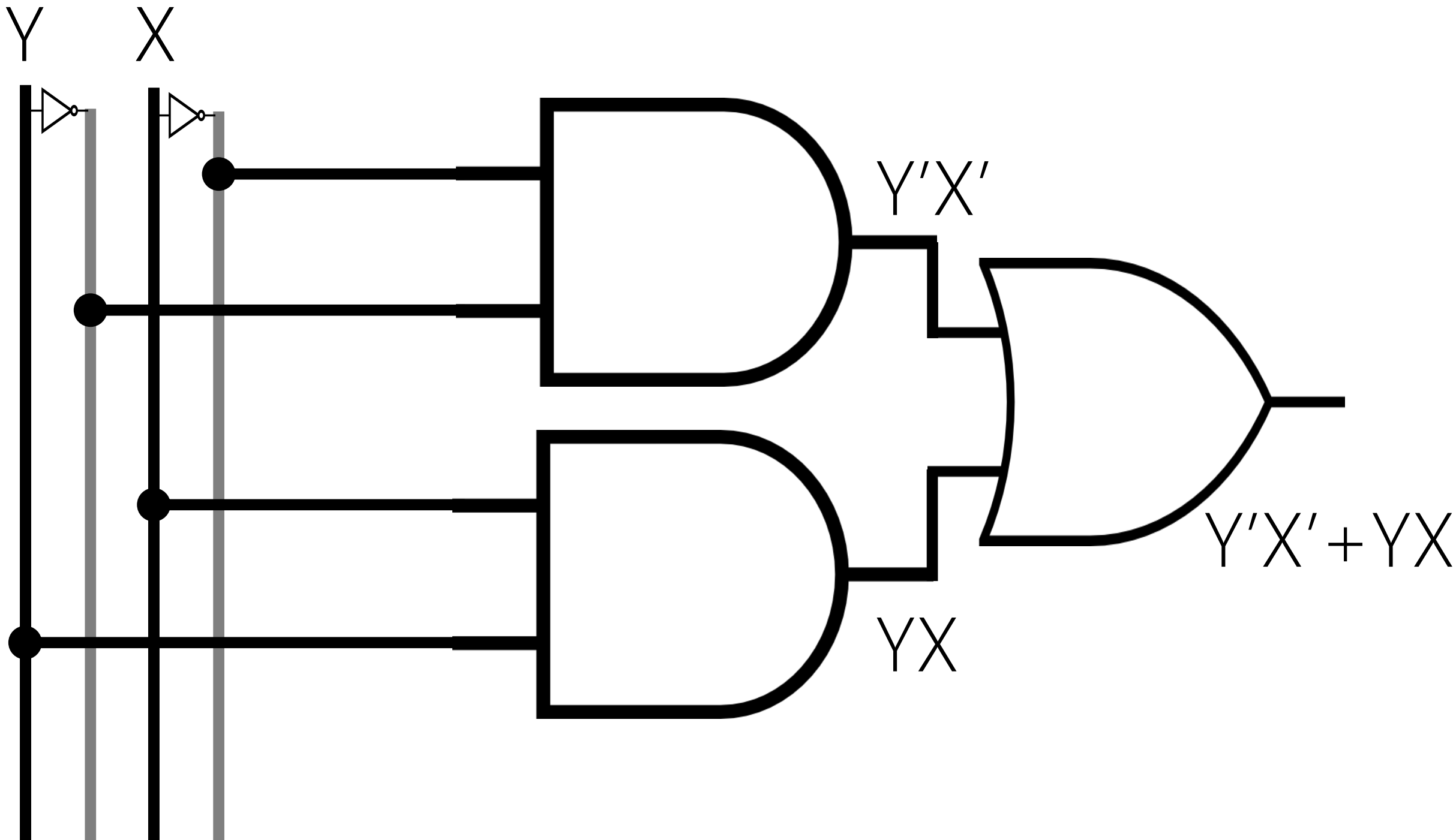
Y	X	$F = F(Y,X) = ?$
0	0	1
0	1	0
1	0	0
1	1	1

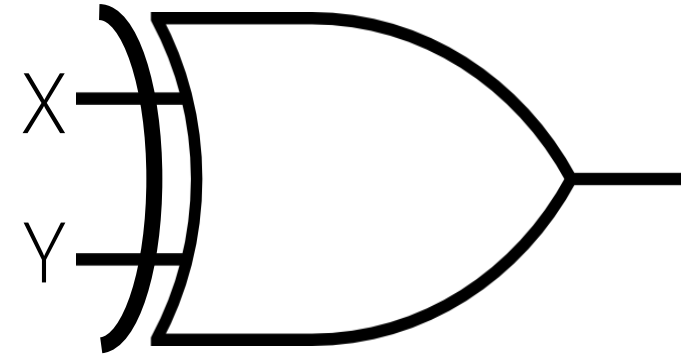
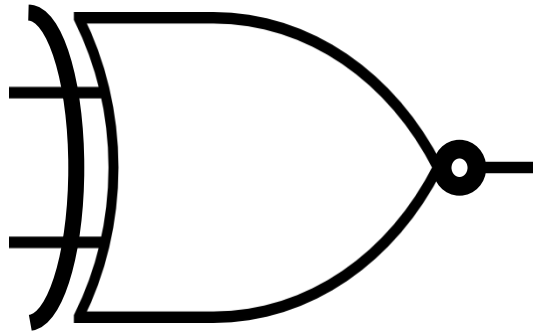
Y	X	$F = F(Y,X) = Y'X'$
0	0	1
0	1	0
1	0	0
1	1	1

Y	X	$F = F(Y,X) = Y'X' + YX$
0	0	1
0	1	0
1	0	0
1	1	1

Y	X	$F = F(Y,X) = m_0 + m_3$
0	0	1
0	1	0
1	0	0
1	1	1

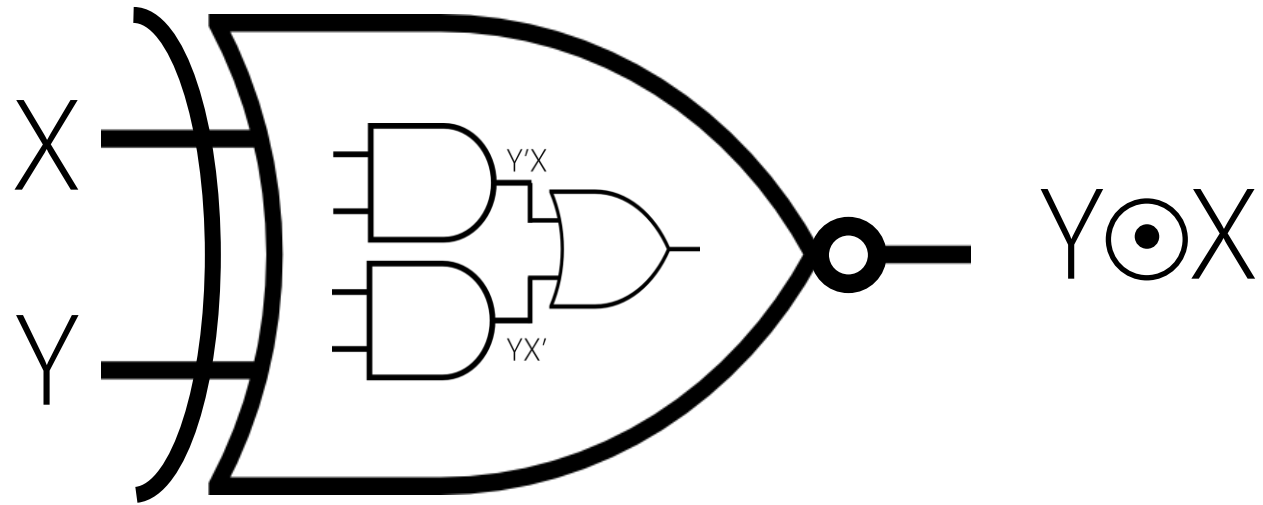
Y	X	$F = F(Y,X) = \sum m(0,3)$
0	0	1
0	1	0
1	0	0
1	1	1





Y	X	$F = F(Y,X) = Y'X' + YX = m_0 + m_3$	$F = F(Y,X) = Y'X + YX' = m_1 + m_2$
0	0	1	0
0	1	0	1
1	0	0	1
1	1	1	0

NOT Exclusive-OR (NXOR $\rightarrow$ XNOR)



Y	X	$F = F(Y,X) = Y'X' + YX = m_0 + m_3$
0	0	1
0	1	0
1	0	0
1	1	1

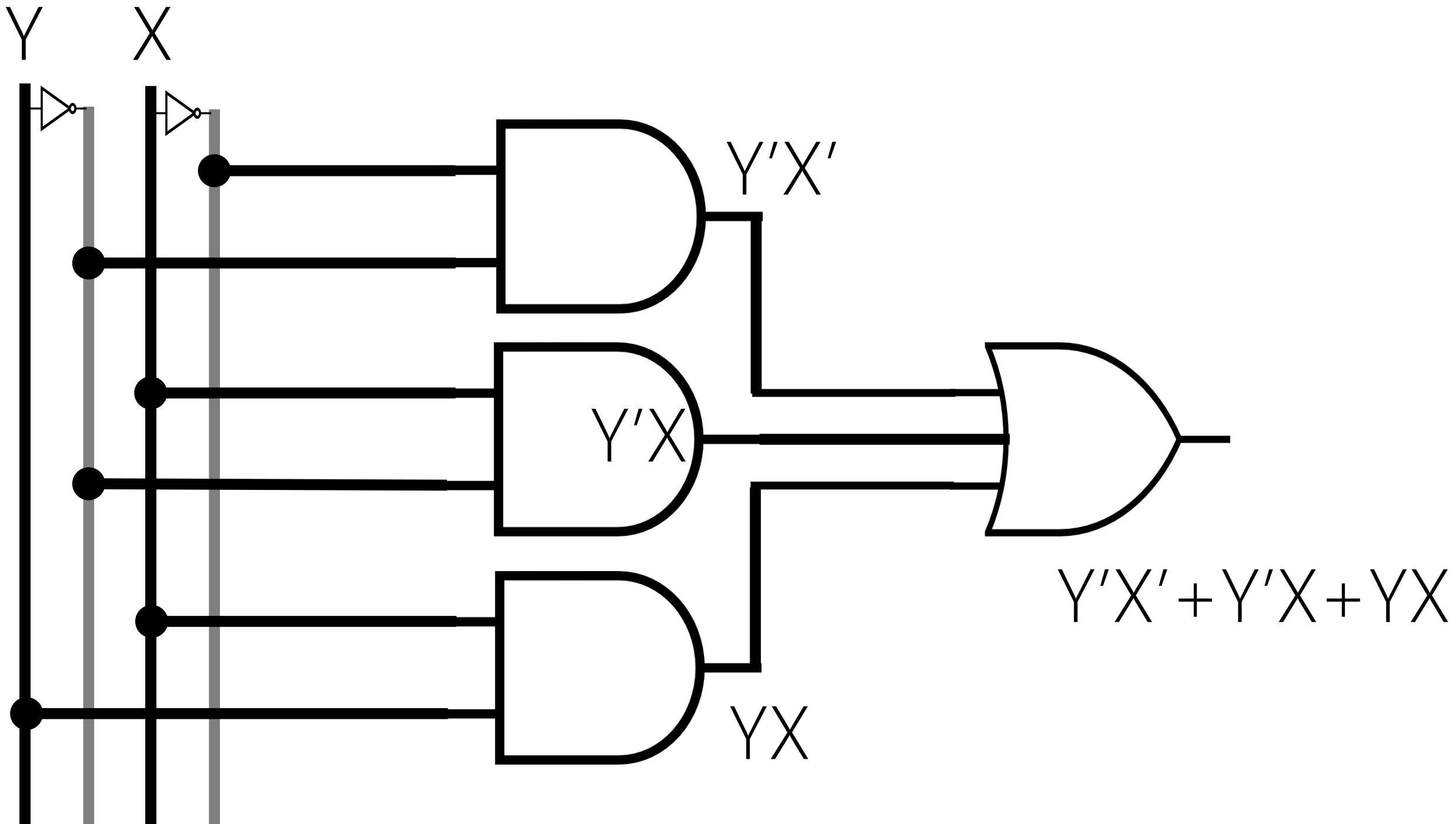
Y	X	$F = F(Y,X) = ?$
0	0	1
0	1	1
1	0	0
1	1	1

Y	X	$F = F(Y,X) = Y'X'$
0	0	1
0	1	1
1	0	0
1	1	1

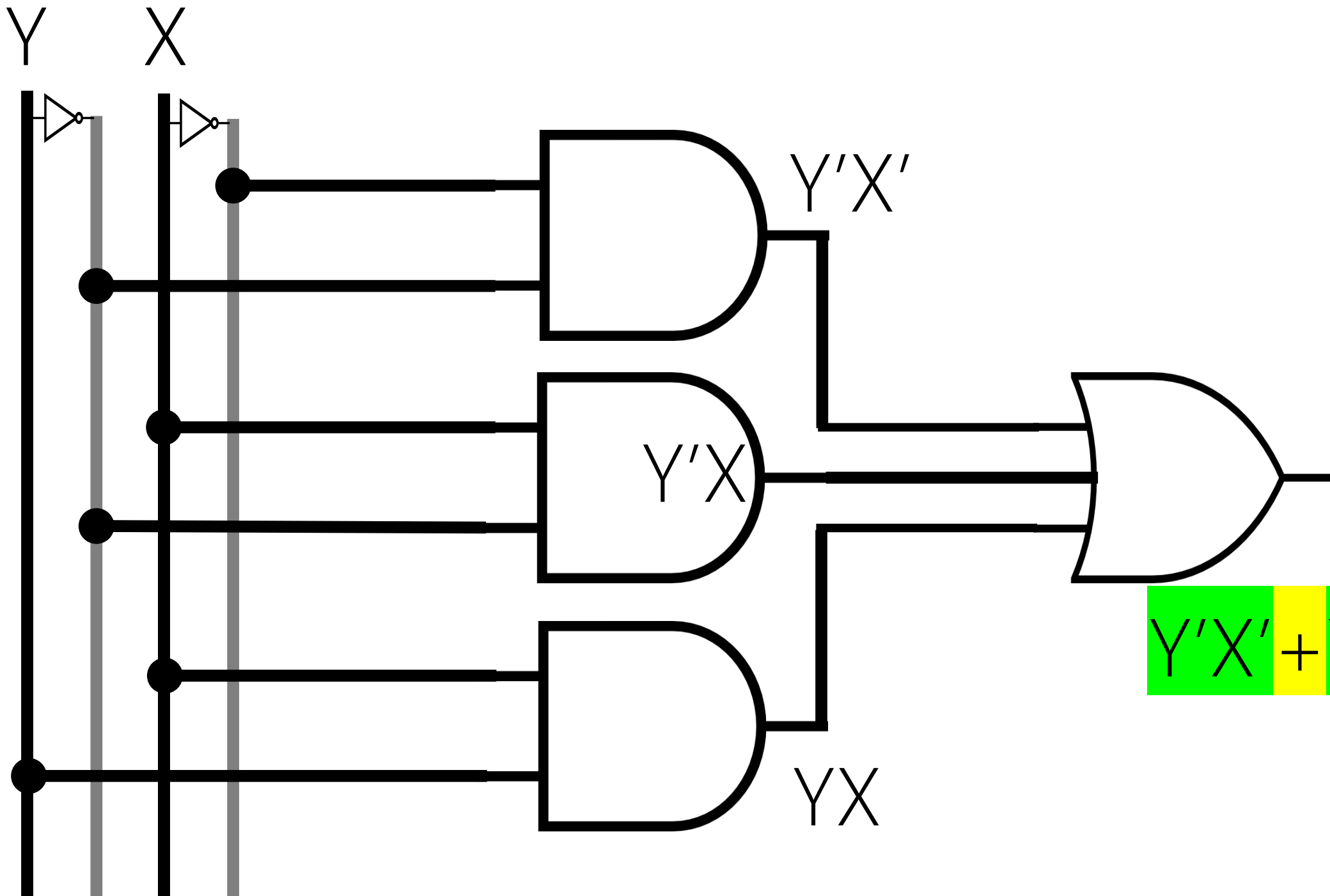
Y	X	$F = F(Y,X) = Y'X' + Y'X$
0	0	1
0	1	1
1	0	0
1	1	1

Y	X	$F = F(Y,X) = Y'X' + Y'X + YX$
0	0	1
0	1	1
1	0	0
1	1	1

Y	X	$F = F(Y, X) = m_0 + m_1 + m_3$ $= \sum m(0, 1, 3)$
0	0	1
0	1	1
1	0	0
1	1	1

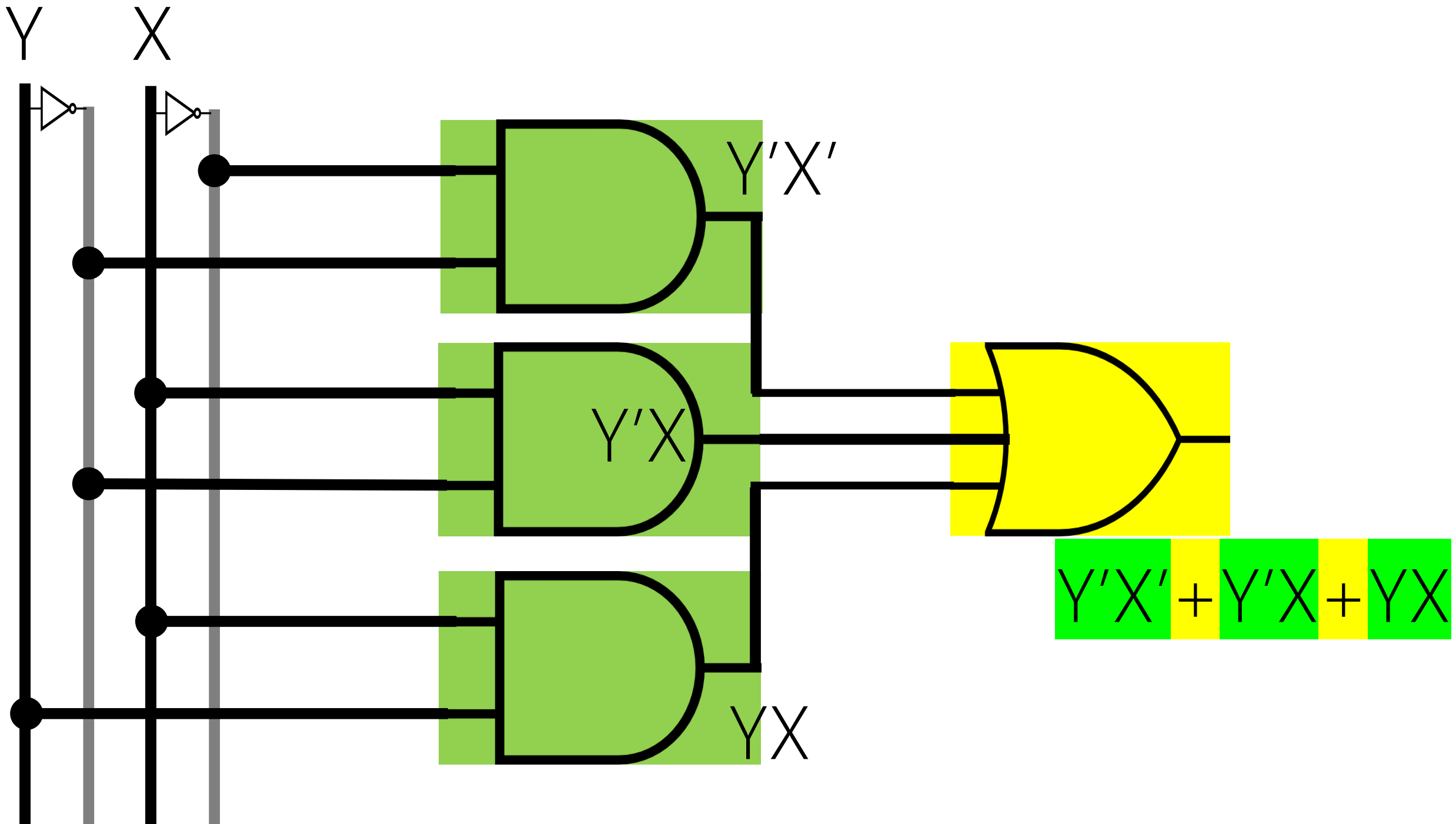


SUM OF PRODUCTS (SOP)



$$Y'X' + Y'X + YX$$

2 LEVELS
AND → OR



Given 3 inputs, design a circuit to determine if there is even number of 1

TRUTH TABLE $\leftrightarrow$ minterm

[illegible][illegible]

Z	Y	X	F(Z,Y,X)=?
0	0	0	?
0	0	1	?
0	1	0	?
0	1	1	?
1	0	0	?
1	0	1	?
1	1	0	?
1	1	1	?

Z	Y	X	F(Z,Y,X)=?
0	0	0	1
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	0

Z	Y	X	$F(Z,Y,X)=Z'Y'X'$
0	0	0	1
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	0

Z	Y	X	$F(Z,Y,X)=Z'Y'X'+Z'YX$
0	0	0	1
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	0

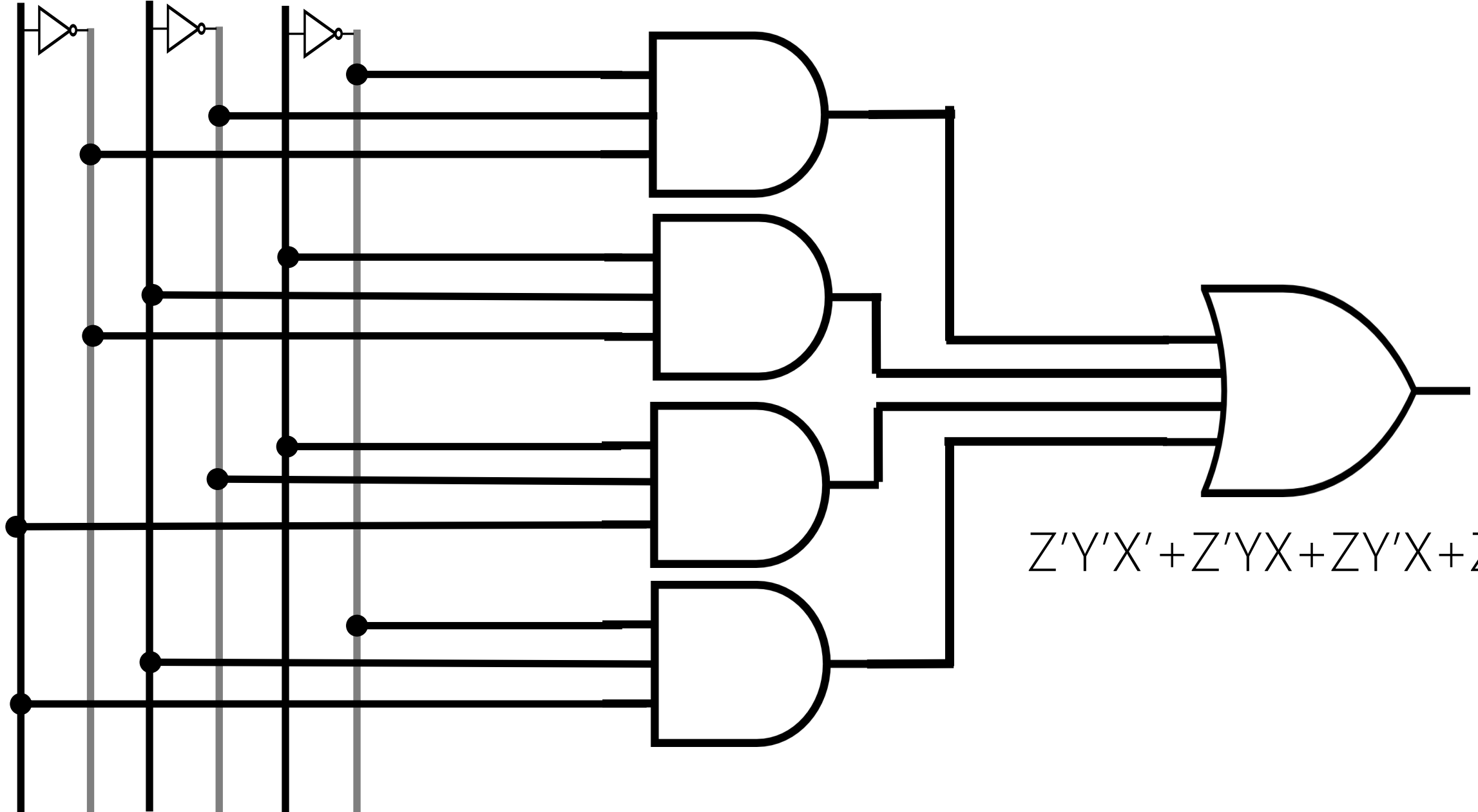
Z	Y	X	$F(Z,Y,X)=Z'Y'X'+Z'YX+ZY'X$
0	0	0	1
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	0

Z	Y	X	$F(Z,Y,X)=Z'Y'X'+Z'YX+ZY'X+ZYX'$
0	0	0	1
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	0

Z	Y	X	$F(Z,Y,X)=m_0+m_3+m_5+m_6$
0	0	0	1
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	0

Z	Y	X	$F(Z,Y,X)=m_0+m_3+m_5+m_6=\sum m(0,3,5,6)$
0	0	0	1
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	0

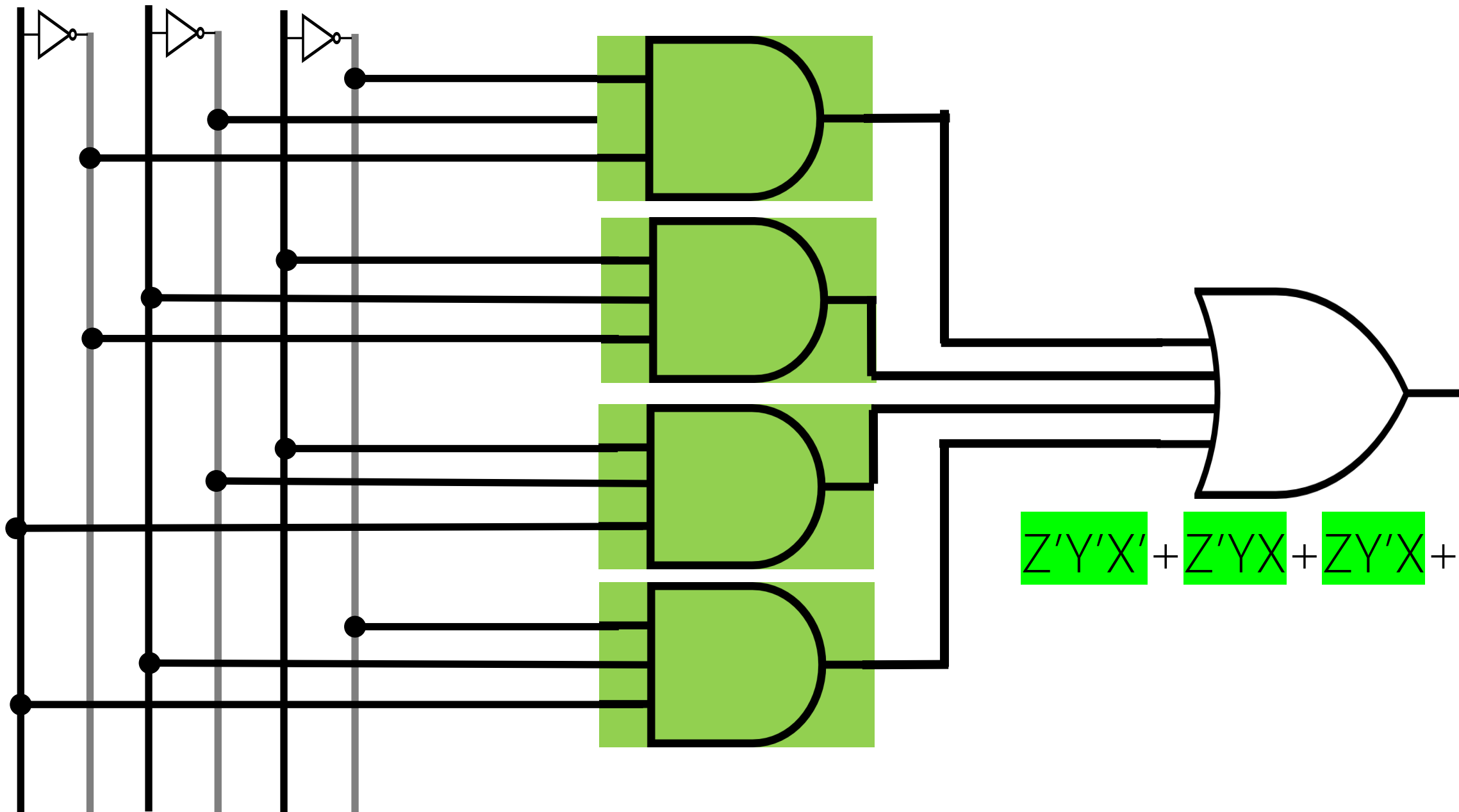
Z Y X



$$Z'Y'X' + Z'YX + ZY'X + ZYX'$$

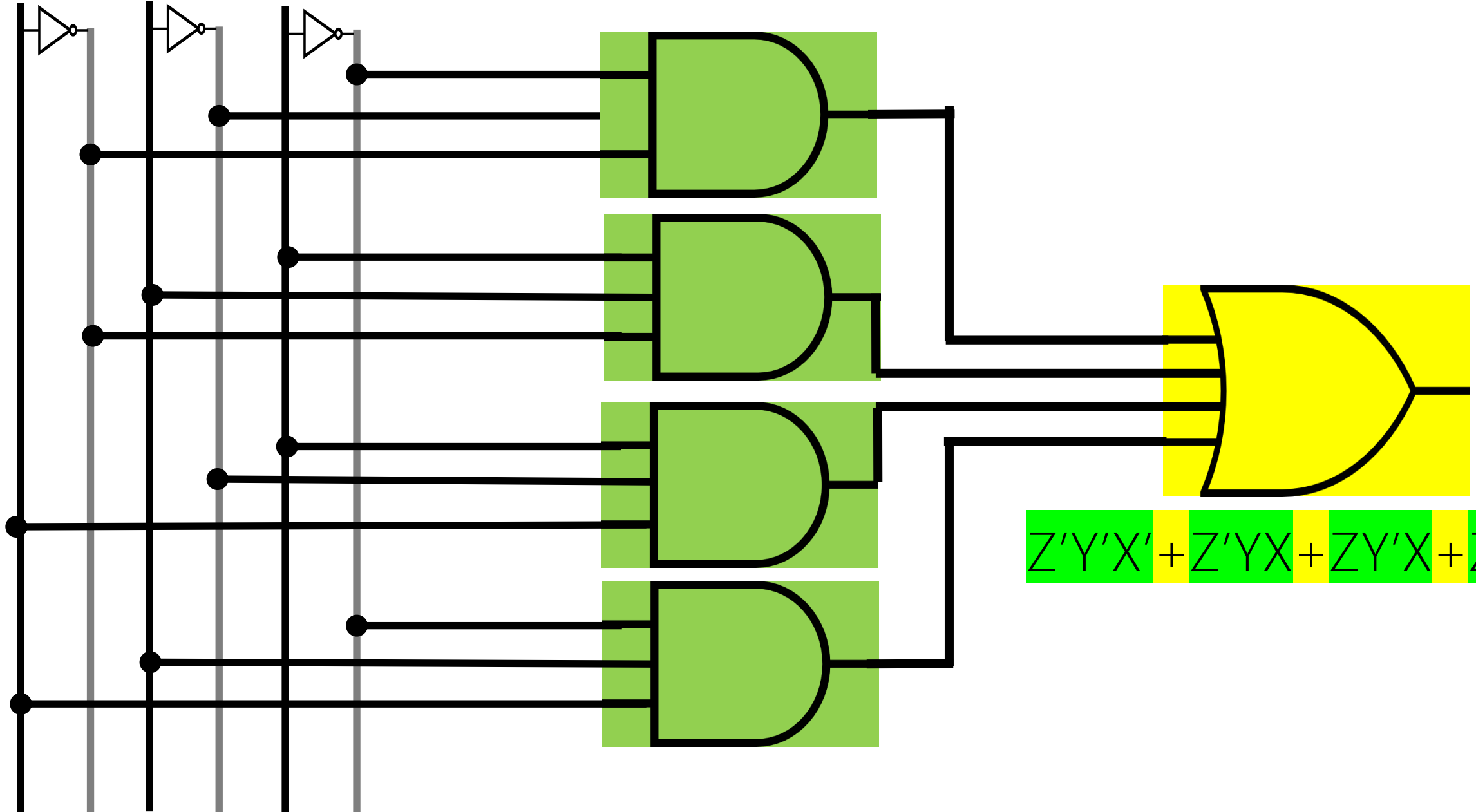
SUM OF PRODUCTS (SOP)
2 LEVELS AND-OR

Z Y X



$$Z'Y'X' + Z'YX + ZY'X + ZYX'$$

Z Y X



$$Z'Y'X' + Z'YX + ZY'X + ZYX'$$

SHOW THE REMAINDER (MOD)
NUMBER % 3 = ?

TRUTH TABLE $\leftrightarrow$ minterm

[illegible][illegible]

WHAT IS THE RANGE OF NUMBERS?

WHAT IS THE RANGE OF NUMBERS?

$$[0, 15]_{10}$$

HOW MANY INPUT **BINARY** VARIABLES?

$$[0, 15]_{10} = [0, 1111]_2 = [0000, 1111]_2$$

W	Z	Y	X
0	0	0	0
0	0	0	1
0	0	1	0
0	0	1	1
0	1	0	0
0	1	0	1
0	1	1	0
0	1	1	1
1	0	0	0
1	0	0	1
1	0	1	0
1	0	1	1
1	1	0	0
1	1	0	1
1	1	1	0
1	1	1	1

WHAT IS THE RANGE OF OUTPUT?

WHAT IS THE RANGE OF OUTPUT?

The remainder of any number divided by 3 is 0, 1, 2

WHAT IS THE RANGE OF OUTPUT?

$$[0, 2]_{10}$$

HOW MANY **BOOLEAN** FUNCTION?

$$[0, 2]_{10} = [0, 10]_2 = [00, 10]_2$$

W	Z	Y	X	F ₁	F ₂
0	0	0	0		
0	0	0	1		
0	0	1	0		
0	0	1	1		
0	1	0	0		
0	1	0	1		
0	1	1	0		
0	1	1	1		
1	0	0	0		
1	0	0	1		
1	0	1	0		
1	0	1	1		
1	1	0	0		
1	1	0	1		
1	1	1	0		
1	1	1	1		

W	Z	Y	X	F ₁	F ₂
0	0	0	0	0	0
0	0	0	1		
0	0	1	0		
0	0	1	1		
0	1	0	0		
0	1	0	1		
0	1	1	0		
0	1	1	1		
1	0	0	0		
1	0	0	1		
1	0	1	0		
1	0	1	1		
1	1	0	0		
1	1	0	1		
1	1	1	0		
1	1	1	1		

W	Z	Y	X	F ₁	F ₂
0	0	0	0	0	0
0	0	0	1	0	1
0	0	1	0		
0	0	1	1		
0	1	0	0		
0	1	0	1		
0	1	1	0		
0	1	1	1		
1	0	0	0		
1	0	0	1		
1	0	1	0		
1	0	1	1		
1	1	0	0		
1	1	0	1		
1	1	1	0		
1	1	1	1		

W	Z	Y	X	F ₁	F ₂
0	0	0	0	0	0
0	0	0	1	0	1
0	0	1	0	1	0
0	0	1	1		
0	1	0	0		
0	1	0	1		
0	1	1	0		
0	1	1	1		
1	0	0	0		
1	0	0	1		
1	0	1	0		
1	0	1	1		
1	1	0	0		
1	1	0	1		
1	1	1	0		
1	1	1	1		

W	Z	Y	X	F ₁	F ₂
0	0	0	0	0	0
0	0	0	1	0	1
0	0	1	0	1	0
0	0	1	1	0	0
0	1	0	0		
0	1	0	1		
0	1	1	0		
0	1	1	1		
1	0	0	0		
1	0	0	1		
1	0	1	0		
1	0	1	1		
1	1	0	0		
1	1	0	1		
1	1	1	0		
1	1	1	1		

W	Z	Y	X	F ₁	F ₂
0	0	0	0	0	0
0	0	0	1	0	1
0	0	1	0	1	0
0	0	1	1	0	0
0	1	0	0	0	1
0	1	0	1	1	0
0	1	1	0	0	0
0	1	1	1	0	1
1	0	0	0	1	0
1	0	0	1	0	0
1	0	1	0	0	1
1	0	1	1	1	0
1	1	0	0	0	0
1	1	0	1	0	1
1	1	1	0	1	0
1	1	1	1	0	0

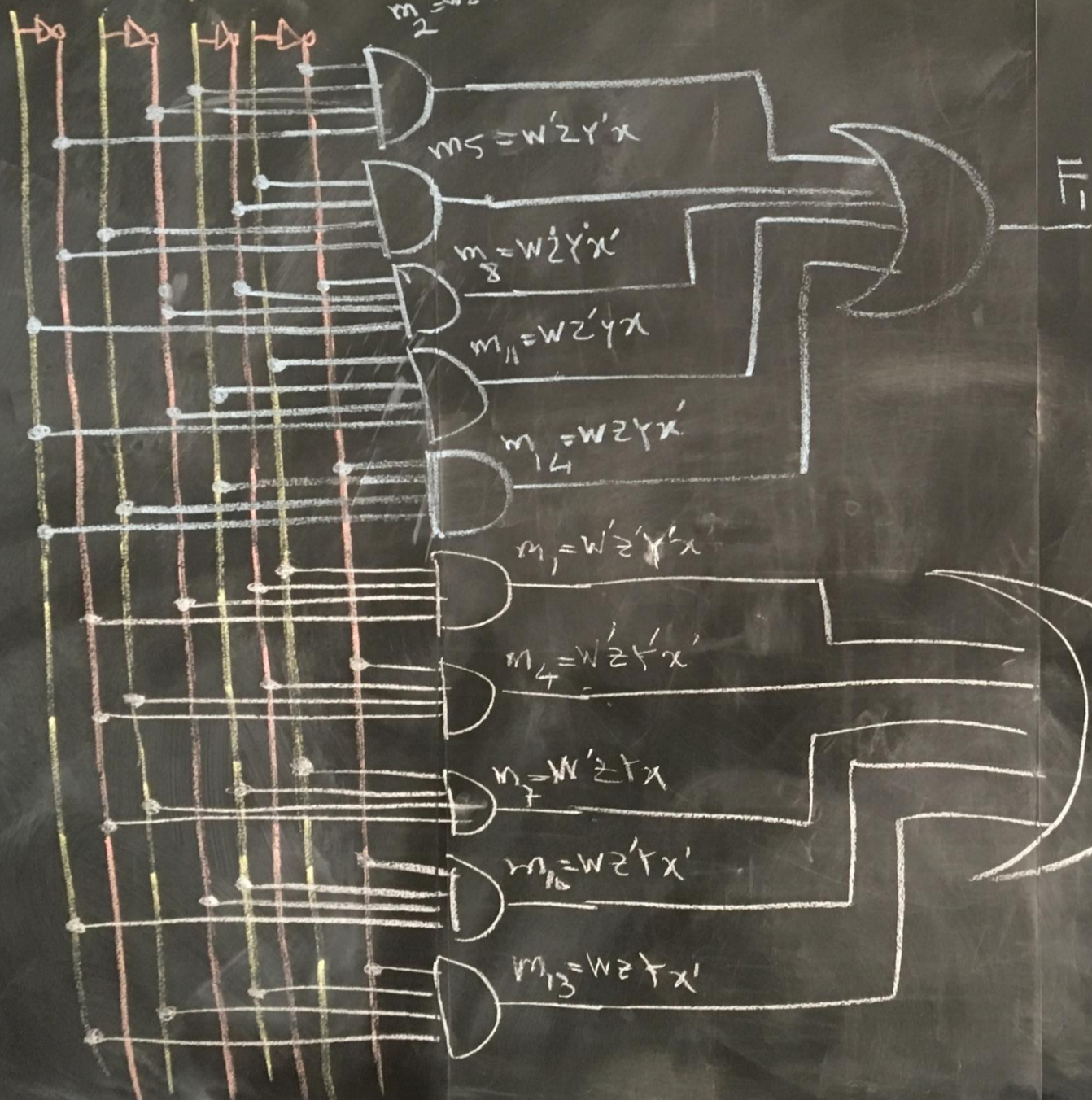
minterms

W	Z	Y	X	$F_1=m_2+m_5+m_8+m_{11}+m_{14}$	F_2
0	0	0	0	0	0
0	0	0	1	0	1
0	0	1	0	1	0
0	0	1	1	0	0
0	1	0	0	0	1
0	1	0	1	1	0
0	1	1	0	0	0
0	1	1	1	0	1
1	0	0	0	1	0
1	0	0	1	0	0
1	0	1	0	0	1
1	0	1	1	1	0
1	1	0	0	0	0
1	1	0	1	0	1
1	1	1	0	1	0
1	1	1	1	0	0

W	Z	Y	X	$F_1 = \sum m(2,5,8,11,14)$	$F_2 = m_1 + m_4 + m_7 + m_{10} + m_{13}$
0	0	0	0	0	0
0	0	0	1	0	1
0	0	1	0	1	0
0	0	1	1	0	0
0	1	0	0	0	1
0	1	0	1	1	0
0	1	1	0	0	0
0	1	1	1	0	1
1	0	0	0	1	0
1	0	0	1	0	0
1	0	1	0	0	1
1	0	1	1	1	0
1	1	0	0	0	0
1	1	0	1	0	1
1	1	1	0	1	0
1	1	1	1	0	0

W	Z	Y	X	$F_1=\sum m(2,5,8,11,14)$	$F_2=\sum m(1,4,7,10,13)$
0	0	0	0	0	0
0	0	0	1	0	1
0	0	1	0	1	0
0	0	1	1	0	0
0	1	0	0	0	1
0	1	0	1	1	0
0	1	1	0	0	0
0	1	1	1	0	1
1	0	0	0	1	0
1	0	0	1	0	0
1	0	1	0	0	1
1	0	1	1	1	0
1	1	0	0	0	0
1	1	0	1	0	1
1	1	1	0	1	0
1	1	1	1	0	0

W Z Y X



$$F_1 = \sum m(2, 5, 8, 11, 14)$$

$$F_2 = \sum m(1, 4, 7, 10, 13)$$

No reuse for minterms